

Module - 5

Hybrid and Transmission Parameters

Dr. Sathya P
Associate Professor
SENSE

Hybrid Parameters

- Impedance and admittance parameters do not exist always in a network.

$$\begin{aligned}V_1 &= h_{11}I_1 + h_{12}V_2 \\ I_2 &= h_{21}I_1 + h_{22}V_2\end{aligned}$$

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = [h] \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

The h terms are known as the *hybrid parameters*

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}, \quad h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0}$$

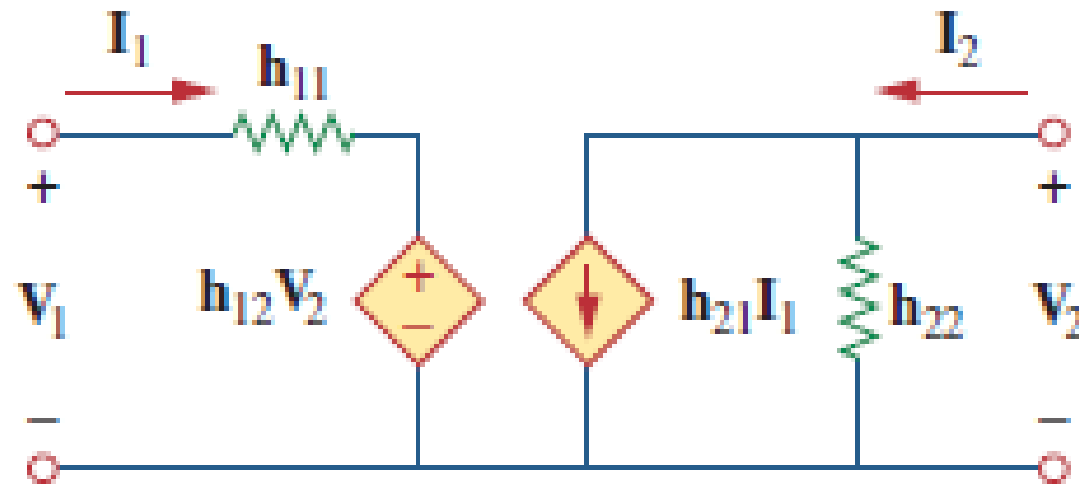
$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}, \quad h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

h_{11} = Short-circuit input impedance

h_{12} = Open-circuit reverse voltage gain

h_{21} = Short-circuit forward current gain

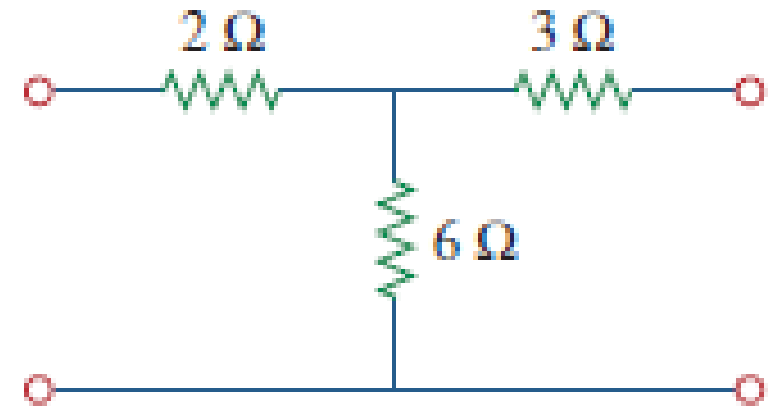
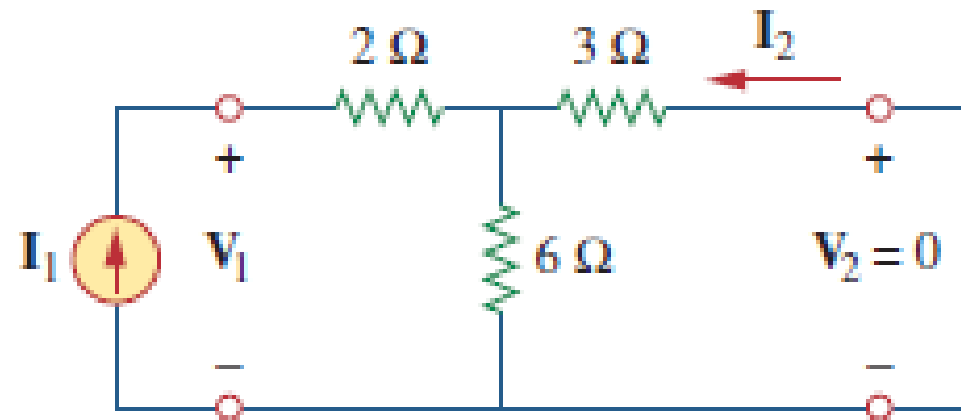
h_{22} = Open-circuit output admittance



Numerical - 1

Find the hybrid parameters for the two-port network

Solution:



$$V_1 = I_1(2 + 3 \parallel 6) = 4I_1$$

$$h_{11} = \frac{V_1}{I_1} = 4 \Omega$$

by current division,
$$-\mathbf{I}_2 = \frac{6}{6 + 3}\mathbf{I}_1 = \frac{2}{3}\mathbf{I}_1$$

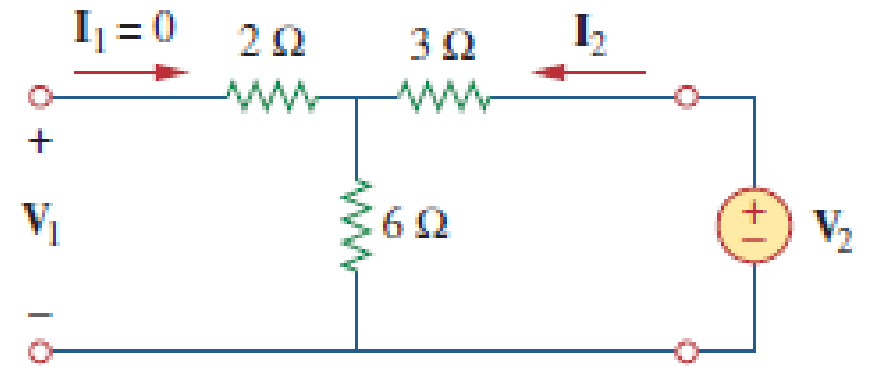
$$\mathbf{h}_{21} = \frac{\mathbf{I}_2}{\mathbf{I}_1} = -\frac{2}{3}$$

$$\mathbf{V}_1 = \frac{6}{6 + 3}\mathbf{V}_2 = \frac{2}{3}\mathbf{V}_2$$

$$\mathbf{h}_{12} = \frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{2}{3}$$

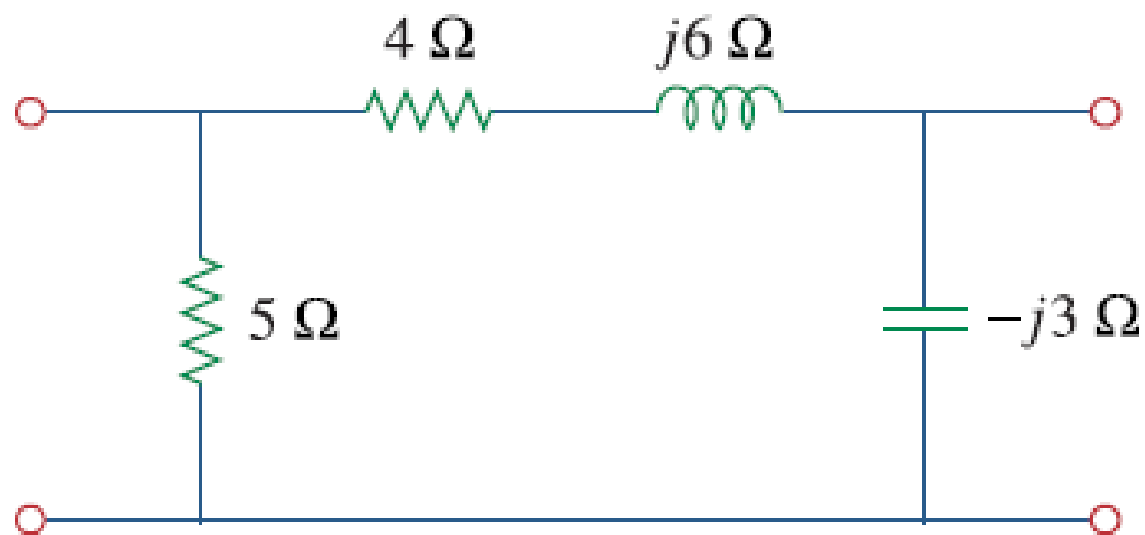
$$\mathbf{V}_2 = (3 + 6)\mathbf{I}_2 = 9\mathbf{I}_2$$

$$\mathbf{h}_{22} = \frac{\mathbf{I}_2}{\mathbf{V}_2} = \frac{1}{9}\mathbf{S}$$

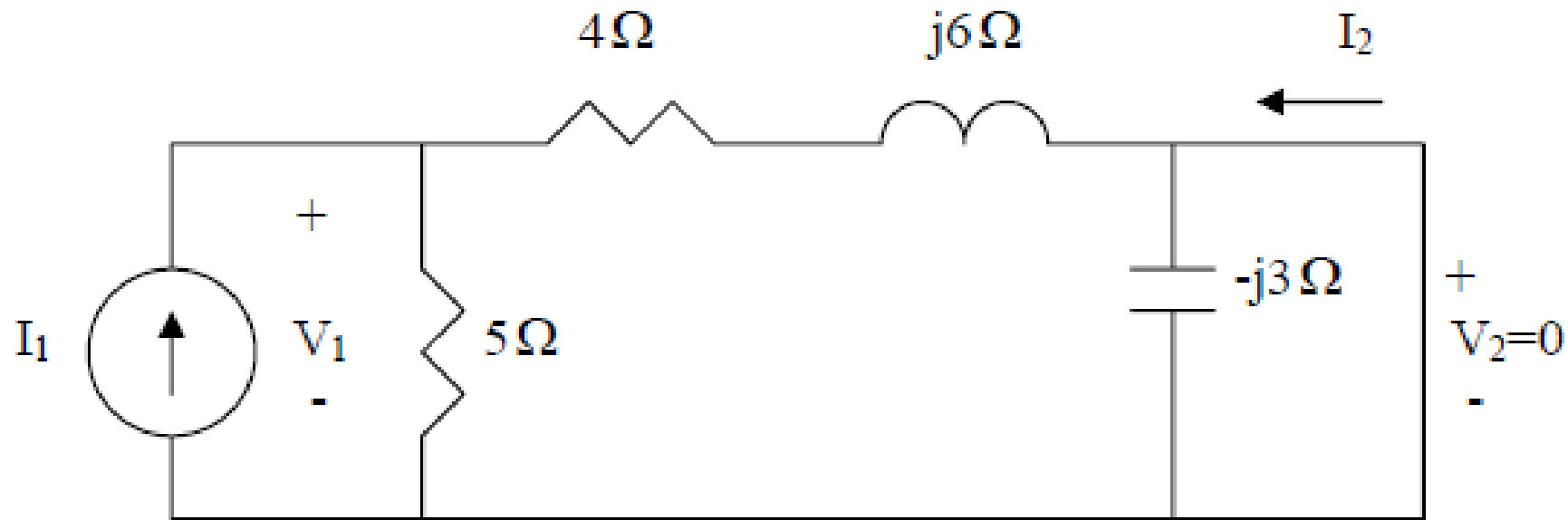


Numerical - 2

Find the hybrid parameters for the two-port network



To get h_{11} and h_{21} , consider the circuit below.



$$V_1 = 5 // (4 + j6) I_1 = \frac{5(4 + j6)I_1}{9 + j6} \quad h_{11} = \frac{V_1}{I_1} = 3.0769 + j1.2821$$

$$\text{Also, } I_2 = -\frac{5}{9 + j6} I_1 \quad \longrightarrow \quad h_{21} = \frac{I_2}{I_1} = -0.3846 + j0.2564$$

To get h_{22} and h_{12} , consider the circuit below.



$$V_1 = \frac{5}{9 + j6} V_2 \quad \longrightarrow \quad h_{12} = \frac{V_1}{V_2} = \frac{5}{9 + j6} = 0.3846 - j0.2564$$

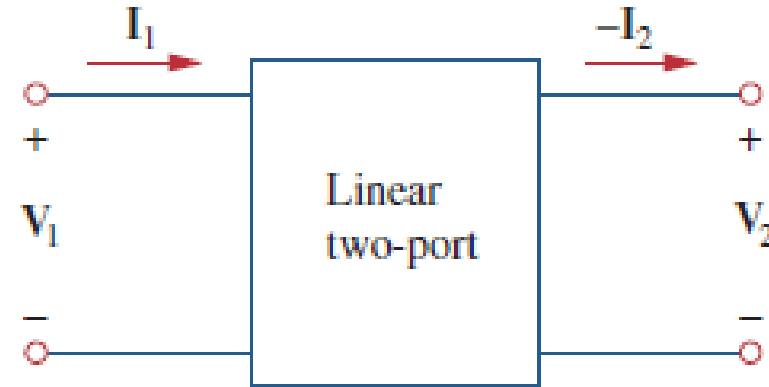
$$V_2 = -j3 \parallel (9 + j6) I_2 \quad \longrightarrow \quad h_{22} = \frac{I_2}{V_2} = \frac{1}{-j3 \parallel (9 + j6)} = \frac{9 + j3}{-j3(9 + j6)} \\ = 0.0769 + j0.2821$$

Thus,

$$[h] = \begin{bmatrix} 3.077 + j1.2821 & 0.3846 - j0.2564 \\ -0.3846 + j0.2564 & 0.0769 + j0.2821 \end{bmatrix}$$

Transmission Parameters

$$\begin{aligned} V_1 &= AV_2 - BI_2 \\ I_1 &= CV_2 - DI_2 \end{aligned}$$



$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} = [T] \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

- This parameter provides the measure of how a circuit transmits voltage and current from a source to a load. They are called as transmission parameters or ABCD parameters and are useful in the analysis of transmission lines (Cables and fibers).
- Used in the design of telephone systems, microwave networks and radars.

The transmission parameters are determined as

$$\begin{aligned} \mathbf{A} &= \left. \frac{\mathbf{V}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0}, & \mathbf{B} &= -\left. \frac{\mathbf{V}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0} \\ \mathbf{C} &= \left. \frac{\mathbf{I}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0}, & \mathbf{D} &= -\left. \frac{\mathbf{I}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0} \end{aligned}$$

Thus, the transmission parameters are called, specifically,

A = Open-circuit voltage ratio

B = Negative short-circuit transfer impedance

C = Open-circuit transfer admittance

D = Negative short-circuit current ratio

Relationship Between Parameters

	z		y		h		T	
z	z₁₁	z₁₂	$\frac{y_{22}}{\Delta_y}$	$-\frac{y_{12}}{\Delta_y}$	$\frac{\Delta_h}{h_{22}}$	$\frac{h_{12}}{h_{22}}$	$\frac{A}{C}$	$\frac{\Delta_T}{C}$
	z₂₁	z₂₂	$-\frac{y_{21}}{\Delta_y}$	$\frac{y_{11}}{\Delta_y}$	$-\frac{h_{21}}{h_{22}}$	$\frac{1}{h_{22}}$	$\frac{1}{C}$	$\frac{D}{C}$
y	$\frac{z_{22}}{\Delta_z}$	$-\frac{z_{12}}{\Delta_z}$	y₁₁	y₁₂	$\frac{1}{h_{11}}$	$-\frac{h_{12}}{h_{11}}$	$\frac{D}{B}$	$-\frac{\Delta_T}{B}$
	$-\frac{z_{21}}{\Delta_z}$	$\frac{z_{11}}{\Delta_z}$	y₂₁	y₂₂	$\frac{h_{21}}{h_{11}}$	$\frac{\Delta_h}{h_{11}}$	$-\frac{1}{B}$	$\frac{A}{B}$
h	$\frac{\Delta_z}{z_{22}}$	$\frac{z_{12}}{z_{22}}$	$\frac{1}{y_{11}}$	$-\frac{y_{12}}{y_{11}}$	h₁₁	h₁₂	$\frac{B}{D}$	$\frac{\Delta_T}{D}$
	$-\frac{z_{21}}{z_{22}}$	$\frac{1}{z_{22}}$	$\frac{y_{21}}{y_{11}}$	$\frac{\Delta_y}{y_{11}}$	h₂₁	h₂₂	$-\frac{1}{D}$	$\frac{C}{D}$
T	$\frac{z_{11}}{z_{21}}$	$\frac{\Delta_z}{z_{21}}$	$-\frac{y_{22}}{y_{21}}$	$-\frac{1}{y_{21}}$	$-\frac{\Delta_h}{h_{21}}$	$-\frac{h_{11}}{h_{21}}$	A	B
	$\frac{1}{z_{21}}$	$\frac{z_{22}}{z_{21}}$	$-\frac{\Delta_y}{y_{21}}$	$-\frac{y_{11}}{y_{21}}$	$-\frac{h_{22}}{h_{21}}$	$-\frac{1}{h_{21}}$	C	D

Numerical - 3

The y parameters of a network are:

$$[y] = \begin{bmatrix} 0.5 & -0.2 \\ -0.2 & 0.4 \end{bmatrix}$$

Determine the z parameters for the network.

Solution

$$z_{11} = \frac{y_{22}}{\Delta y} = \frac{0.4}{0.16} = 2.5, \quad \Delta y = y_{11}y_{22} - y_{21}y_{12} = 0.5 \times 0.4 - 0.2 \times 0.2 = 0.16$$

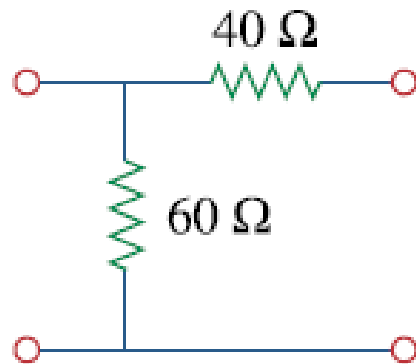
$$z_{12} = \frac{-y_{12}}{\Delta y} = \frac{0.2}{0.16} = 1.25 = z_{21}$$

$$z_{22} = \frac{y_{11}}{\Delta y} = \frac{0.5}{0.16} = 3.125$$

$$[z] = \underline{\underline{\begin{bmatrix} 2.5 & 1.25 \\ 1.25 & 3.125 \end{bmatrix} \Omega}}$$

Numerical - 4

Find the h parameters for the network



$$\Delta_z = z_{11} z_{22} - z_{12} z_{21} = 6000 - 3600 = 2400$$

$$h_{11} = \frac{\Delta_z}{z_{22}} = \frac{2400}{100} = 24, \quad h_{12} = \frac{z_{12}}{z_{22}} = \frac{60}{100} = 0.6$$

$$h_{21} = \frac{-z_{21}}{z_{22}} = -0.6, \quad h_{22} = \frac{1}{z_{22}} = 0.01$$

Solution

Convert to z parameters; then, convert to h parameters

$$z_{11} = z_{12} = z_{21} = 60 \, \Omega, \quad z_{22} = 100 \, \Omega$$

$$\underline{[h] = \begin{bmatrix} 24 \, \Omega & 0.6 \\ -0.6 & 0.01 \, \text{S} \end{bmatrix}}$$